1. Write a C++ program to calculate Simple Interest and Compound Interest.

```cpp
#include<iostream>
#include<cmath>
using namespace std;

int main()
{
   float p, r, t, SI, CI;

   cout<<"Enter Principal: ";
   cin>>p;

   cout<<"Enter Rate: ";
   cin>>r;

   cout<<"Enter Time: ";
   cin>>t;

   SI = (p * r * t) / 100;

   CI = p * pow((1 + r/100), t) - p;

   cout<<"Simple Interest = "<<SI<<endl;
   cout<<"Compound Interest = "<<CI<<endl;

   return 0;
}
```

```
PROBLEMS    OUTPUT    TERMINAL    PORTS    DEBUG CONSOLE

~/Documents/CDAC/C++/doc/SLA1
g++ SICI.cpp

~/Documents/CDAC/C++/doc/SLA1
./a.out
Enter Principal: 1000
Enter Rate: 10
Enter Time: 2
Simple Interest = 200
Compound Interest = 210
```

2. Write a C++ Program to transpose one matrix and display both matrices with a proper prompt.

```
#include<iostream>
using namespace std;

int main()
{
   int a[3][3], t[3][3];

   cout<<"Enter 3x3 matrix:\n";
   for(int i=0;i<3;i++)
       for(int j=0;j<3;j++)
           cin>>a[i][j];

   for(int i=0;i<3;i++)
       for(int j=0;j<3;j++)
           t[j][i] = a[i][j];

   cout<<"Original Matrix:\n";
   for(int i=0;i<3;i++){
       for(int j=0;j<3;j++)
           cout<<a[i][j]<<" ";
       cout<<endl;
   }

   cout<<"Transpose Matrix:\n";
   for(int i=0;i<3;i++){
       for(int j=0;j<3;j++)
           cout<<t[i][j]<<" ";
       cout<<endl;
   }

   return 0;
}
```

```
PROBLEMS    OUTPUT    TERMINAL    PORTS    DEBUG CONSOLE

~/Documents/CDAC/C++/doc/SLA1
g++ matrix.cpp

~/Documents/CDAC/C++/doc/SLA1
./a.out
Enter 3x3 matrix:
234
124
456
347
2345
234
234
788
543
Original Matrix:
234 124 456
347 2345 234
234 788 543
Transpose Matrix:
234 347 234
124 2345 788
456 234 543
```

3. Write a C++ program to process students' marks with the help of classes. The class has private variables for name, mark 1, mark 2, mark 3. It has two member functions – getdata( ) – to get input and result( ) – to print the results. All subjects mark must be >= 50 for pass otherwise fail. Implement this program for n number of students.

```
#include<iostream>
using namespace std;

class Student
{
private:
   string name;
   int m1, m2, m3;

public:
   void getdata()
   {
       cout<<"Enter Name: ";
       cin>>name;

       cout<<"Enter 3 Marks: ";
```

```
        cin>>m1>>m2>>m3;
    }

    void result()
    {
        if(m1>=50 && m2>=50 && m3>=50)
            cout<<name<<" : PASS"<<endl;
        else
            cout<<name<<" : FAIL"<<endl;
    }
};

int main()
{
    int n;
    cout<<"Enter number of students: ";
    cin>>n;

    Student s[n];

    for(int i=0;i<n;i++)
        s[i].getdata();

    for(int i=0;i<n;i++)
        s[i].result();

    return 0;
}
```

PROBLEMS OUTPUT TERMINAL PORTS DEBUG CONSOLE

```
clear
Focus folder in explorer (cmd + click)
~/Documents/CDAC/C++/doc/SLA1
g++ passfail.cpp

~/Documents/CDAC/C++/doc/SLA1
./a.out
Enter number of students: 5
Enter Name: Ashu
Enter 3 Marks: 65
87
94
Enter Name: Amit
Enter 3 Marks: 49
42
65
Enter Name: Amar
Enter 3 Marks: 48
58
50
Enter Name: Arun
Enter 3 Marks: 45
65
34
Enter Name: Akash
Enter 3 Marks: 45
98
34
Ashu : PASS
Amit : FAIL
Amar : FAIL
Arun : FAIL
Akash : FAIL
```

4. Write a C++ program that allows the user to input 5 grades, i.e. marks between 0 - 100. The program must calculate the average mark, and state the number of marks less than 65.

```
#include<iostream>
using namespace std;

int main()
{
   int marks[5], count=0;
   float sum=0;

   cout<<"Enter 5 marks:\n";

   for(int i=0;i<5;i++)
   {
       cin>>marks[i];
       sum += marks[i];

       if(marks[i] < 65)
           count++;
   }

   cout<<"Average = "<<sum/5<<endl;
   cout<<"Marks less than 65 = "<<count<<endl;

   return 0;
}
```

```
~/Documents/CDAC/C++/doc/SLA1
g++ average.cpp

~/Documents/CDAC/C++/doc/SLA1
./a.out
Enter 5 marks:
78
56
65
98
88
Average = 77
Marks less than 65 = 1

~/Documents/CDAC/C++/doc/SLA1
```

5. Write a menu driven C++ program to add, subtract and multiply given two matrices defined in class. Modify your program using operator overloading method.

```
#include<iostream>
using namespace std;

class Matrix
{
public:
   int a[2][2];

   void input()
   {
       for(int i=0;i<2;i++)
           for(int j=0;j<2;j++)
               cin>>a[i][j];
   }

   void display()
   {
       for(int i=0;i<2;i++){
           for(int j=0;j<2;j++)
               cout<<a[i][j]<<" ";
           cout<<endl;
       }
   }

   Matrix operator+(Matrix m)
   {
       Matrix r;
       for(int i=0;i<2;i++)
           for(int j=0;j<2;j++)
               r.a[i][j]=a[i][j]+m.a[i][j];
       return r;
   }

   Matrix operator-(Matrix m)
   {
       Matrix r;
       for(int i=0;i<2;i++)
           for(int j=0;j<2;j++)
               r.a[i][j]=a[i][j]-m.a[i][j];
       return r;
```

```
    }

    Matrix operator*(Matrix m)
    {
        Matrix r;
        for(int i=0;i<2;i++)
            for(int j=0;j<2;j++){
                r.a[i][j]=0;
                for(int k=0;k<2;k++)
                    r.a[i][j]+=a[i][k]*m.a[k][j];
            }
        return r;
    }
};

int main()
{
    Matrix m1,m2,m3;
    int choice;

    cout<<"Enter Matrix 1:\n";
    m1.input();

    cout<<"Enter Matrix 2:\n";
    m2.input();

    cout<<"1.Add\n2.Subtract\n3.Multiply\n";
    cin>>choice;

    switch(choice)
    {
        case 1: m3 = m1 + m2; break;
        case 2: m3 = m1 - m2; break;
        case 3: m3 = m1 * m2; break;
    }

    cout<<"Result:\n";
    m3.display();
}
```

```
PROBLEMS    OUTPUT    TERMINAL    PORTS    DEBUG

~/Documents/CDAC/C++/doc/SLA1
g++ operatoroverloading.cpp

~/Documents/CDAC/C++/doc/SLA1
./a.out
Enter Matrix 1:
45
43
12
54
Enter Matrix 2:
89
65
47
90
1.Add
2.Subtract
3.Multiply
1
Result:
134 108
59 144
```

6. Create a class DONOR that contains donor number, donor name, age address, sex and blood group. Write a Menu driven C++ program to display the number, name and address of the donors for the following categories:
a. blood donors having the blood group O+
b. blood donors in the age group between 18 to 25
c. female donors having blood group A in the age between 21 and 24.

```
#include <iostream>
using namespace std;

class DONOR
{
   int dno, age;
   char name[30], address[50], sex[10], bg[5];

public:
   void accept()
   {
       cout<<"Enter Donor No: ";
       cin>>dno;

       cout<<"Enter Name: ";
       cin>>name;

       cout<<"Enter Age: ";
       cin>>age;

       cout<<"Enter Address: ";
       cin>>address;

       cout<<"Enter Sex: ";
       cin>>sex;

       cout<<"Enter Blood Group: ";
       cin>>bg;
   }

   void display()
   {
       cout<<"Donor No: "<<dno<<endl;
       cout<<"Name: "<<name<<endl;
       cout<<"Address: "<<address<<endl;
       cout<<"-----------------"<<endl;
   }

   char* getBG()
   {
       return bg;
   }
```

```
   int getAge()
   {
       return age;
   }

   char* getSex()
   {
       return sex;
   }
};

int main()
{
   DONOR d[10];
   int n, ch;

   cout<<"Enter number of donors: ";
   cin>>n;

   for(int i=0;i<n;i++)
   {
       d[i].accept();
   }

   do
   {
       cout<<"\n1. Blood Group O+";
       cout<<"\n2. Age between 18 to 25";
       cout<<"\n3. Female donors with Blood Group A and Age 21-24";
       cout<<"\n4. Exit";
       cout<<"\nEnter choice: ";
       cin>>ch;

       switch(ch)
       {
           case 1:
               cout<<"\nDonors with Blood Group O+\n";
               for(int i=0;i<n;i++)
               {
                   if(strcmp(d[i].getBG(),"O+")==0)
                       d[i].display();
```

```
            }
            break;

        case 2:
            cout<<"\nDonors Age 18-25\n";
            for(int i=0;i<n;i++)
            {
                if(d[i].getAge()>=18 && d[i].getAge()<=25)
                    d[i].display();
            }
            break;

        case 3:
            cout<<"\nFemale donors Blood Group A Age 21-24\n";
            for(int i=0;i<n;i++)
            {
                if(strcmp(d[i].getSex(),"Female")==0 &&
                   strcmp(d[i].getBG(),"A")==0 &&
                   d[i].getAge()>=21 && d[i].getAge()<=24)
                    d[i].display();
            }
            break;

    }

}while(ch!=4);

return 0;
}
```

```
~/Documents/CDAC/C++/doc/SLA1
g++ donor.cpp

~/Documents/CDAC/C++/doc/SLA1
./a.out
Enter number of donors: 2
Enter Donor No: 1
Enter Name: Ajit
Enter Age: 24
Enter Address: Pune
Enter Sex: Male
Enter Blood Group: O+
Enter Donor No: 2
Enter Name: Nikita
Enter Age: 22
Enter Address: Pune
Enter Sex: Female
Enter Blood Group: A

1. Blood Group O+
2. Age between 18 to 25
3. Female donors with Blood Group A and Age 21-24
4. Exit
Enter choice: 3

Female donors Blood Group A Age 21-24
Donor No: 2
Name: Nikita
Address: Pune
-----------------

1. Blood Group O+
2. Age between 18 to 25
3. Female donors with Blood Group A and Age 21-24
4. Exit
Enter choice: 1

Donors with Blood Group O+
Donor No: 1
Name: Ajit
Address: Pune
-----------------

1. Blood Group O+
2. Age between 18 to 25
3. Female donors with Blood Group A and Age 21-24
4. Exit
Enter choice: 4
```

7. An electricity board charges the following rates to domestic users:
For the first 100 units – ₹. 5.5P per unit
For next 200 units – ₹. 7.25P per unit
Beyond 300 units – ₹. 8.75P per unit

All users are charged a minimum of ₹. 60.

If the total amount is more than ₹. 750 then an additional surcharge of 10% is added on the higher slab units.

Write a C++ program to read the names of users and number of units consumed and print out the charges with names.

```
#include <iostream>
using namespace std;

int main()
{
   char name[50];
   int units;
   float amount = 0, surcharge = 0;

   cout << "Enter User Name: ";
   cin >> name;

   cout << "Enter Units Consumed: ";
   cin >> units;

   if(units <= 100)
   {
       amount = units * 5.5;
   }
   else if(units <= 300)
   {
       amount = (100 * 5.5) + (units - 100) * 7.25;
   }
   else
   {
       amount = (100 * 5.5) + (200 * 7.25) + (units - 300) * 8.75;
   }

   if(amount < 60)
   {
```

```
        amount = 60;
    }

    if(amount > 750)
    {
        surcharge = amount * 0.10;
        amount = amount + surcharge;
    }

    cout << "\nUser Name: " << name;
    cout << "\nUnits Consumed: " << units;
    cout << "\nTotal Bill Amount: Rs. " << amount << endl;

    return 0;
}
```

```
PROBLEMS    OUTPUT    TERMINAL    PORTS    DEBUG CON

~/Documents/CDAC/C++/doc/SLA1
g++ electricitycharges.cpp

~/Documents/CDAC/C++/doc/SLA1
./a.out
Enter User Name: Ajit
Enter Units Consumed: 563

User Name: Ajit
Units Consumed: 563
Total Bill Amount: Rs. 4731.38
```

8.Implement the calculator through a C++ program (use switch). The program should ask the user to input two operands and an operator. Your program should print the result of performing the required operation on the input operands as the following output:

Enter operand 1: 5
Enter operand 2: 10
Enter operation: /

The result of 5/10 is 0.5

```
#include <iostream>

using namespace std;


int main()

{

   float op1, op2, result;

   char oper;


   cout << "Enter operand 1: ";

   cin >> op1;


   cout << "Enter operand 2: ";

   cin >> op2;


   cout << "Enter operation (+, -, *, /): ";

   cin >> oper;


   switch(oper)

   {

       case '+':

           result = op1 + op2;

           break;


       case '-':

           result = op1 - op2;

           break;


       case '*':

           result = op1 * op2;

           break;
```

```
        case '/':
            result = op1 / op2;
            break;

        default:
            cout << "Invalid operator";
            return 0;
    }

    cout << "The result of " << op1 << oper << op2 << " is " << result;

    return 0;
}
```

```
~/Documents/CDAC/C++/doc/SLA1
g++ calculator.cpp

~/Documents/CDAC/C++/doc/SLA1
./a.out
Enter operand 1: 12
Enter operand 2: 65
Enter operation (+, -, *, /): +
The result of 12+65 is 77%

~/Documents/CDAC/C++/doc/SLA1
```

9. Create Time Class which contains the members as hour, minute and second. Write a C++ program that accepts member data for one Time class object from the keyboard. Then ask the user to perform one of the following functions on that Time object:

- AddHours
- AddMinutes
- AddSeconds

And finally display the newly modified Time information.

```
#include <iostream>

using namespace std;



class Time

{

   int hour, minute, second;



public:

   void accept()

   {

       cout << "Enter Hours: ";

       cin >> hour;



       cout << "Enter Minutes: ";

       cin >> minute;



       cout << "Enter Seconds: ";
```

```
    cin >> second;
}

void addHours(int h)
{
    hour = hour + h;
}

void addMinutes(int m)
{
    minute = minute + m;

    if(minute >= 60)
    {
        hour = hour + minute / 60;
        minute = minute % 60;
    }
}

void addSeconds(int s)
{
    second = second + s;

    if(second >= 60)
```

```
        {
            minute = minute + second / 60;
            second = second % 60;
        }

        if(minute >= 60)
        {
            hour = hour + minute / 60;
            minute = minute % 60;
        }
    }

    void display()
    {
        cout << "Modified Time: ";
        cout << hour << ":" << minute << ":" << second << endl;
    }
};

int main()
{
    Time t;
    int choice, value;
```

```
t.accept();


cout << "\n1. Add Hours";
cout << "\n2. Add Minutes";
cout << "\n3. Add Seconds";
cout << "\nEnter Choice: ";
cin >> choice;


cout << "Enter value to add: ";
cin >> value;


switch(choice)
{
    case 1:
        t.addHours(value);
        break;


    case 2:
        t.addMinutes(value);
        break;


    case 3:
        t.addSeconds(value);
        break;
```

```
        default:
            cout << "Invalid Choice";
    }

    t.display();

    return 0;
}
```

```
PROBLEMS    OUTPUT    TERMINAL    PORTS    DEBUG CON

~/Documents/CDAC/C++/doc/SLA1
g++ timeclass.cpp

~/Documents/CDAC/C++/doc/SLA1
./a.out
Enter Hours: 5
Enter Minutes: 34
Enter Seconds: 43

1. Add Hours
2. Add Minutes
3. Add Seconds
Enter Choice: 1
Enter value to add: 5
Modified Time: 10:34:43
```

10. Write a menu-driven program to perform the following options using separate functions. Accept a number and then do the following operations on this number:

1. Prime number or not
2. Odd or Even
3. Positive or Negative
4. Exit

```
#include <iostream>

using namespace std;



void checkPrime(int n)

{

   int i, flag = 0;



   if(n <= 1)

   {

       cout << "Not a Prime Number\n";

       return;

   }



   for(i = 2; i <= n/2; i++)

   {

       if(n % i == 0)

       {

           flag = 1;

           break;

       }
```

```
    }

    if(flag == 0)
        cout << "Prime Number\n";
    else
        cout << "Not a Prime Number\n";
}

void checkOddEven(int n)
{
    if(n % 2 == 0)
        cout << "Even Number\n";
    else
        cout << "Odd Number\n";
}

void checkPositiveNegative(int n)
{
    if(n > 0)
        cout << "Positive Number\n";
    else if(n < 0)
        cout << "Negative Number\n";
    else
        cout << "Number is Zero\n";
```

```
}


int main()

{

   int num, choice;


   cout << "Enter a number: ";

   cin >> num;


   do

   {

       cout << "\n--- MENU ---";

       cout << "\n1. Prime number or not";

       cout << "\n2. Odd or Even";

       cout << "\n3. Positive or Negative";

       cout << "\n4. Exit";

       cout << "\nEnter choice: ";

       cin >> choice;


       switch(choice)

       {

           case 1:

               checkPrime(num);

               break;
```

```
        case 2:
            checkOddEven(num);
            break;

        case 3:
            checkPositiveNegative(num);
            break;

        case 4:
            cout << "Exiting Program\n";
            break;

        default:
            cout << "Invalid Choice\n";
    }

  } while(choice != 4);

  return 0;
}
```

```
~/Documents/CDAC/C++/doc/SLA1
g++ number.cpp

~/Documents/CDAC/C++/doc/SLA1
./a.out
Enter a number: 54

--- MENU ---
1. Prime number or not
2. Odd or Even
3. Positive or Negative
4. Exit
Enter choice: 1
Not a Prime Number

--- MENU ---
1. Prime number or not
2. Odd or Even
3. Positive or Negative
4. Exit
Enter choice: 4
Exiting Program
```

11. Write a function reverse() that reverses the character strings. Use it to write a program that reverses its input a line at a time.

```
#include <iostream>

#include <cstring>

using namespace std;



void reverse(char str[])
```

```
{
   int i, len;
   char temp;

   len = strlen(str);

   for(i = 0; i < len/2; i++)
   {
       temp = str[i];
       str[i] = str[len-1-i];
       str[len-1-i] = temp;
   }
}

int main()
{
   char str[100];

   cout << "Enter a line: ";
   cin.getline(str, 100);

   reverse(str);

   cout << "Reversed string: " << str << endl;
```

```
    return 0;

}
```

```
PROBLEMS    OUTPUT    TERMINAL    PORTS

~/Documents/CDAC/C++/doc/SLA1
g++ reverse.cpp

~/Documents/CDAC/C++/doc/SLA1
./a.out
Enter a line: HELO
Reversed string: OLEH
```

12. Write a C++ Program to Compare Two given Strings and display the result.

```
#include <iostream>

#include <cstring>

using namespace std;



int main()

{

    char str1[50], str2[50];
```

```
    cout << "Enter first string: ";

    cin >> str1;



    cout << "Enter second string: ";

    cin >> str2;



    if(strcmp(str1, str2) == 0)

        cout << "Both strings are equal.";

    else

        cout << "Strings are not equal.";



    return 0;

}
```

```
~/Documents/CDAC/C++/doc/SLA1
g++ comparetwostrings.cpp

~/Documents/CDAC/C++/doc/SLA1
./a.out
Enter first string: STRING
Enter second string: GOWITHFLOW
Strings are not equal.%

~/Documents/CDAC/C++/doc/SLA1
```

13. Write a program in C++ to determine how many of the characters are vowels and how many are consonants in a given line of text. Also, terminate the string when the input character encountered is other than the alphabet (a-z or A-Z) and blank spaces.

Hint:

a. When the input string is "C++ FOR BEGINNERS, TEST YOUR C++ PROGRAMMING STRENGTHS":
The program will consider the string "C++ for beginners" only because ',' is encountered.

b. When the input string is "C++ LEARNING IS NOT A PROBLEM":
Program will consider the character "C" only because the '+' is encountered.

```
#include <iostream>

using namespace std;



int main()

{

   char str[200];

   int vowels = 0, consonants = 0;



   cout << "Enter a line: ";

   cin.getline(str, 200);



   for(int i = 0; str[i] != '\0'; i++)

   {

       if((str[i] >= 'a' && str[i] <= 'z') ||

          (str[i] >= 'A' && str[i] <= 'Z'))
```

```
        {
            if(str[i]=='a'||str[i]=='e'||str[i]=='i'||str[i]=='o'||str[i]=='u'||
               str[i]=='A'||str[i]=='E'||str[i]=='I'||str[i]=='O'||str[i]=='U')
                vowels++;
            else
                consonants++;
        }
        else if(str[i] == ' ')
        {
            continue;
        }
        else
        {
            break;
        }
    }

    cout << "Vowels = " << vowels << endl;
    cout << "Consonants = " << consonants << endl;

    return 0;
}
```

```
PROBLEMS    OUTPUT    TERMINAL    PORTS    DEBUG

~/Documents/CDAC/C++/doc/SLA1
g++ vowelconsonant.cpp

~/Documents/CDAC/C++/doc/SLA1
./a.out
Enter a line: Hello World
Vowels = 3
Consonants = 7

~/Documents/CDAC/C++/doc/SLA1
```

14. Create a C++ class for a stock item abstract data type. It should have the attributes of stock level (an integer) and unit price (a float). Define the methods to return the values of these two attributes and to set them using parameters. Add two more methods to allow stock receipt and issues updating the stock level as appropriate. Write a menu driven C++ program to solve the problem.

```
#include <iostream>

using namespace std;



class Stock
```

```
{
   int stockLevel;
   float unitPrice;

public:

   void setValues(int s, float p)
   {
       stockLevel = s;
       unitPrice = p;
   }

   int getStock()
   {
       return stockLevel;
   }

   float getPrice()
   {
       return unitPrice;
   }

   void receiveStock(int qty)
   {
```

```
        stockLevel = stockLevel + qty;

        cout << "Stock received successfully\n";

    }



    void issueStock(int qty)

    {

        if(qty <= stockLevel)

        {

            stockLevel = stockLevel - qty;

            cout << "Stock issued successfully\n";

        }

        else

        {

            cout << "Insufficient stock\n";

        }

    }

};



int main()

{

    Stock s;

    int choice, qty, stock;

    float price;
```

```
cout << "Enter Initial Stock Level: ";

cin >> stock;



cout << "Enter Unit Price: ";

cin >> price;



s.setValues(stock, price);



do

{

    cout << "\n--- MENU ---";

    cout << "\n1. View Stock";

    cout << "\n2. Receive Stock";

    cout << "\n3. Issue Stock";

    cout << "\n4. Exit";

    cout << "\nEnter choice: ";

    cin >> choice;



    switch(choice)

    {

        case 1:

            cout << "Current Stock: " << s.getStock() << endl;

            cout << "Unit Price: " << s.getPrice() << endl;

            break;
```

```
        case 2:
            cout << "Enter quantity to receive: ";
            cin >> qty;
            s.receiveStock(qty);
            break;

        case 3:
            cout << "Enter quantity to issue: ";
            cin >> qty;
            s.issueStock(qty);
            break;

    }

  } while(choice != 4);

  return 0;
}
```

PROBLEMS OUTPUT TERMINAL PORTS

```
└ g++ stock.cpp

~/Documents/CDAC/C++/doc/SLA1
└ ./a.out
Enter Initial Stock Level: 72
Enter Unit Price: 5

--- MENU ---
1. View Stock
2. Receive Stock
3. Issue Stock
4. Exit
Enter choice: 1
Current Stock: 72
Unit Price: 5

--- MENU ---
1. View Stock
2. Receive Stock
3. Issue Stock
4. Exit
Enter choice: 4
```

15. Raising a number n to power p is the same as multiplying by itself p times. Write a function called power() that takes a double value for n and an int value for p and returns the result as double value. Use default argument of 2 for p, so that if this argument is omitted, the number will be squared. Write a main() function that gets values from the user to test this function.

```
#include <iostream>

using namespace std;



double power(double n, int p = 2)

{

   double result = 1;



   for(int i = 1; i <= p; i++)

   {

       result = result * n;

   }



   return result;

}



int main()

{

   double num;

   int p;
```

```
    cout << "Enter number: ";

    cin >> num;



    cout << "Enter power (enter 0 if you want default square): ";

    cin >> p;



    if(p == 0)

        cout << "Result = " << power(num) << endl;

    else

        cout << "Result = " << power(num, p) << endl;



    return 0;

}
```

```
PROBLEMS    OUTPUT    TERMINAL    PORTS    DEBUG CONSOLE

~/Documents/CDAC/C++/doc/SLA1
g++ power.cpp

~/Documents/CDAC/C++/doc/SLA1
./a.out
Enter number: 5
Enter power (enter 0 if you want default square): 3
Result = 125

~/Documents/CDAC/C++/doc/SLA1
```